



Tidal Zone Prints

Grade 6
Printing



Los Altos School District • 201 Covington Road, Los Altos, CA 94024 • 650.947.1195
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Last Revised: February 2024

Tidal Zone Prints

Length: 90 minutes

Number of Docents: 2

Purpose:

- To introduce students to another method of printmaking
- To give students the opportunity to create their own designs
- To encourage students to look at nature closely and discover the beauty and patterns therein.

Academic Support: Science curriculum study of ocean ecosystems

Visual Arts Standards:

- Formulate an artistic investigation of personally relevant content for creating art. (6.VA:Cr1.2)
- Demonstrate openness in trying new ideas, materials, methods, and approaches in making works of art and design. (6.VA:Cr2.1)
- Reflect on whether personal artwork conveys the intended meaning and revise accordingly. (6.VA:Cr3)

Unit Contains: Supply box and portfolio

Materials docent will bring:

Per student

- 4" square linoleum printing block (sent via Pony)
- 1" x 2" linoleum practice printing block (sent via Pony)
- Six 6" x 9" sheets white drawing paper for the prints (210 total sheets)
- Set of cutting tools (blue handle with round-tip, red handle with V-tip)
- Wooden spoon
- HB pencil
- 6" square sheet of Thai Unryu paper

To share

- Brayers
- Acrylic inking plates
- Printing inks (yellow, magenta, turquoise and silver)
- Spray bottle (for re-hydrating inks when thick or dry)
- Scrubbing brush (for clean-up)
- Toothpicks and Q-tips
- Board showing texture, patterns and illustrations of sea life
- Samples
- Reproductions

Materials teacher will provide:Per student

- One 8.5" x 11" sheet of white paper for sketching designs
- Non-mechanical pencils for sketching
- Paper towels
- Paper to cover the inking table(s)

Featured works:

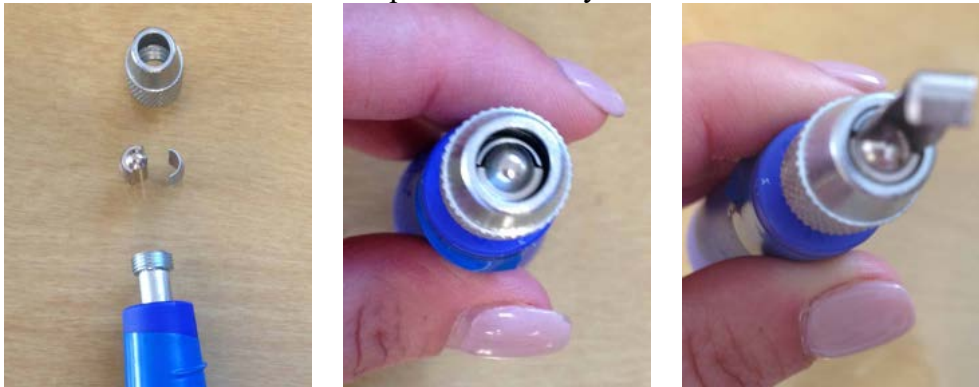
<i>The Great Wave Off Kanagawa</i> (1820–31)	Katsushika Hokusai
<i>Marin Headlands for Land's End</i> (1990)	Tom Killion
<i>Big Sur Spring Sunset</i> (1990)	Tom Killion
<i>A Carp Leaping in a Pool</i>	Katsushika Taito
Poster with various prints	Wayne Thiebaud

Room setup:

Write your name and assistant's name on the board.

Students will work at their desks except for inking. Give each student sketch paper, a non-mechanical pencil if they don't have one, print paper, a paper towel, a wooden spoon, a 4" printing block, a 1" x 2" printing block, and a set of carving tools.

Make sure that the carving tool hand chucks holding the blades in place are tightened all the way so the blades do not come out. There are replacement blades in the unit if you find that the blades are dull and in need of changing. To change a blade, slightly loosen the hand chuck and pull out the blade. Insert a new blade between the two pieces in the hand chuck and tighten. You DO NOT need to unscrew the hand chuck all the way but it's pictured here so you can see the pieces. Please encourage students NOT to unscrew the hand chuck as the interior pieces are easily lost when loose.



For the inking table(s), put down paper if available and then set out at least (8) acrylic inking plates with ink, brayers, Q-tips, toothpicks, and a spray bottle. Each color will appear on 2–3 plates depending on available space. An assisting docent, teacher or parent helper will be in charge of adding more ink to the inking plates as needed. Explain to him/her the technique of inking.

INTRODUCTION

Explain that students will be making prints of creatures that inhabit the coastal tidal area they have been or will be studying.

What is printmaking?

Printmaking is an **art medium** where the artwork is **produced by using a block or a plate** (metal, plastic, wood, silk screen, or linoleum) that has been **altered** by scratching, carving, painting or applying acids and chemicals **to create a design on its surface**. The printing plate is **inked** to transfer the design to paper. By applying **pressure** from an object called a press, the **ink is forced onto the paper** to make a print.

There are numerous different types of prints: serigraphs, woodcuts, etchings, lithographs, etc. (*Quickly show or have on display prints*)



The Great Wave Off Kanagawa (1820–31)
Katsushika Hokusai
Woodcut print
15" x 10"



Marin Headlands for Land's End (1990)
Tom Killion
Wood and linocut print
19" x 12"

OR



A Carp Leaping in a Pool
Katsushika Taito
Woodcut print



Big Sur Spring Sunset (1990)
Tom Killion
Wood and linocut print
19" x 13"



Glass Candy Sticks (1980) Watercolor and colored pencil over lithograph, 21" x 17"



Jawbreaker Machine (1990) Watercolor, pastel and gouache over linocut, 22" x 17"



Sugar Cones (1985) Watercolor over etching, 4" x 5"



Lemon Meringue (1985) Watercolor over etching, 4" x 5"
Wayne Thiebaud

Printing allows the artist to make duplicate copies, either exactly the same or altered in some manner. When the artist prints several prints exactly alike, it is called an **edition**. When a unique image can only be made once, it is called a **monoprint**. A print made on a smooth plate is called a **monotype**.

Prints you will make

You are going to make prints from a linoleum block that has been cut in **relief**, that is, part of the block has been carved or removed to leave a design on a raised surface. Ink is then applied to the design to make a print.

A woodcut is one type of relief print. **Traditionally, the Japanese carved several woodblocks to make one print, one block, for each color.** These were carefully printed one at a time on the same piece of paper to make the final print. Artists like Van Gogh, as well as contemporary printmakers, also used and continue to use this method. Today many artists carve one block to print the basic design and then hand color the paper with paint, pencil, etc., to create individual, unique images. They may also experiment by adding objects like bits of paper and string during the printing process.

You are going to carve just one block, making variations of the design by trying some different techniques. We will not be making an edition. We will be making **monoprints**.

DEMONSTRATION

1. Creating a design on a piece of paper

Display boards showing tide pool inhabitants, samples of prints, and different types of patterns. Explain the process, demonstrating on board when necessary.

- Think of tide pools and the sea for inspiration and **choose one inhabitant** (starfish, kelp, anemone, crab, limpet, snail, mussel) to use as your main subject.
- Make a **few quick sketches** and then draw your final design in a space created by tracing the outline of the linoleum block.
- When drawing your design, you do not have to fill in all the details. **Just suggest ideas.** Details can be completed when you transfer the design to the block or while carving.
- Think about the overall composition. Keep the subject **large** and simple. A strong, simple graphic is easier to carve and will give better results when printing.
- Show its environment. Decide which areas will be **positive and negative**, or dark against light. The **outside edge** of the design will appear next to the white paper.
- **Alternative designs** may be realistic or stylized, delicate or bold, have a plain subject with a textured background, have bold areas of outline, graded areas radiating outwards from the subject, or have varying directional lines.

- **Avoid small curved shapes and tiny details** since they are difficult to carve.
- To avoid confusion while carving, you may lightly **shade the areas that will print dark**. Do this on your sketches.

2. Achieving detailed design or texture

Discuss types of lines and shapes to achieve design and texture: curvy, thick or thin, cross-hatching, gouging, etc.

- Think of lines that resemble the **movement** of water or flow of plants, the **designs** on kelp and the edge of the anemone, **repetition** of whorls on a snail, and **overlapping** of plant leaves.
- Take care to keep **enough space between the lines** so that paint will not fill in the area and lose detail.
- Keep in mind that **lines and textures will print slightly larger** than the cut you have made. Tiny details will not reproduce.

3. Transferring

- Transfer the design to the block by tracing your design at least three times with a non-mechanical pencil. Extra pencils are available in the bin if students do not have them. Place your printing block over the pencil design, invert it so that the paper is on the top and rub with the wooden spoon. Using the edge of the spoon rather than the flat back over the lines is often more effective. The graphite will transfer to the block and any faint lines may be touched up with the pencil.
- Wooden spoons and pencils may be collected after designs are transferred.

4. Carving

Demonstrate how to cut. Show samples and have them practice on their small blocks.

Different tools are used for different lines. The red-handled cutter has a **V-shaped blade for cutting thin lines**. It is better for fine details.

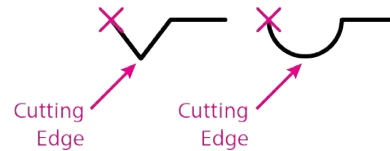
"V" Tool Creates
Narrow Lines



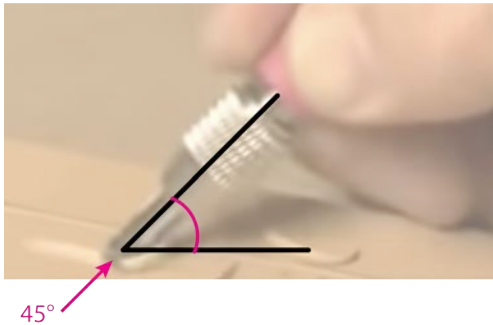
"C" Tool Creates
Wider Lines



The blue-handled cutter has a **C-shaped blade for cutting wide lines**. It is better for removing a large surface area.



Pick up either cutter with your dominant hand and examine the cutting blade. Look carefully and observe that the "V" or "C" has the sharp cutting edge on one side only, and it is not at the tip! The **cutting edge** is actually the scoop of the "C", or the valley of the "V"—NOT the tip/outer edge of the tool. Identify which part of the blade is the sharp cutting edge.



Whether you hold the handle like a flashlight or a pen, make sure you are angling the sharp cutting edge into the block. Place this cutting edge on the linoleum block and angle it at **45-degrees** in relation to the surface. With light, GENTLE pressure, slide the cutting edge of the tool into the linoleum block as if you are slipping a scoop into a carton of ice cream, and pull the tool to create your line.

Practice with different amounts of pressure and finding the correct angle. There should be little resistance—the tool should glide easily through the linoleum and you should be able to see the outer tip of the blade as you carve. If you dig too deeply and cannot see the tip, then the linoleum will remain attached to the block on that side of the blade. Too much pressure or holding it at an incorrect angle makes it difficult to pull the tool.

Use long **continuous strokes**, and turn the block rather than turning your hand.

The carving tools are sharp so **take care** when carving. Keep the hand that is holding the block out of the way of your carving tool at all times.

You may want to **cut the outline of your subject first**. Then cut details within the lines, between the lines, and any large open areas or background that will have texture after your outline is complete.

Also, consider **filling in empty spaces** around your main design with texture for a more interesting result.

Dots (for bubbles or patterns on your creatures) may be carved by slipping the carving tool into the surface of the block and then spinning the block beneath your tool.

Your carved **grooves should be deep enough** for you to feel them when you run your fingertip over the block. If your cuts are too shallow for you to feel, your design will be very faint and hard to see when you make your prints.

NOTE for lefties (and righties)... Remind students to hold their tools at a 45-degree angle. This makes it easier to see their line, which in turn makes carving easier. It can also help for them to position their arms away from their bodies a little instead of pulling directly back towards themselves.

Some lefties may find it is easier to pull the tools away from them so that they maintain the 45-degree angle. Have them experiment on their practice block first to find the technique that works best for them.

Use your fingertips to **remove pieces** that are cut away from the block.

Remember which objects are in **front** and which are **behind**.

5. Printing

When the majority of students are done carving, or close to done, have the class take a break and demonstrate the printing process. Demonstrate printing a single-color print and another using two colors of ink as well as a toothpick and a Q-tip. Also demonstrate what happens when you try to make a second print without re-inking your block.

- Put a dab of ink on an acrylic inking plate and roll in each direction until smooth.
- Apply ink to a carved block using a brayer with a thin layer of ink. Use enough ink to cover the surface, but do not make it so thick that it appears bumpy or uneven.
- Re-ink the brayer as necessary, and re-hydrate the ink on the acrylic plate with water from the spray bottle if the ink is too thick or if it starts drying out.
- Roll the brayer in both directions over your block.
- Place a sheet of 6" x 9" print paper on top of your block (you can fold the paper to create a card or leave flat). Gently press the paper to the block making sure to rub all areas that touch the block.
- To change the color of the ink on your block for your next print, you may add the next color directly over the last color of ink, or print a ghost print on scrap paper to remove the previous ink, or gently wipe the block clean with a damp towel. If you wipe the block, do not make it too wet or ink will flow into the narrow cuts the next time you add ink to the block.

6. Types of prints to be printed

Show samples and explain the printing techniques the students will use. Request that students make at least four prints. They may make these prints in any order they choose. Encourage students to have fun and experiment.

- Make a print using the lightest color ink (yellow). When dry, you can add **additional color with colored pencils** after the ink dries. Think of contrasting colors used next to each other.
- Print at least one **multi-colored print** using any of the following options.
 - Ink your block with a light color (yellow) and print. Re-ink the block with one or more colors and print, carefully aligning (**registering**) the block to the existing print. Re-ink and print as many times as you wish.
 - Ink the block, **then alter the print by adding or subtracting color** with a Q-tip. (Imagine a triad of secondary colors, a split of complementary bright colors, or colors of the sea.)
 - Apply ink to the block by **rolling different colored inks** in three separate areas.
- Ink with a dark color. **Add more detail by removing (scratching)** some of the ink off of the block with a toothpick before making your print. Remove an adequate amount of ink so the lines will show when printed. Scratch lightly as you do not want to permanently mar the surface of your block. Work quickly so the ink does not dry out before you make your print.
- Print one of the prints on colored **unryu paper**. You may use a combination of colors, a single color, or multi-colored layers. Consider which ink will provide the best contrast with the color of the paper.

7. Signature

When the paper is dry, **sign your name** under the print on the **right side**.

Note: If an artist produces identical editions—which we are not doing today—you will see a **date** and **number/number** written on the **left side** under the print. The numbers represent the sequence in which the prints were made, *i.e.*, “32/100” means it is print number 32 out of a total of 100 prints. The first prints, those printed in the experimental stage will read **Artist’s Proof**.

8. Docent Clean-up

- Wash all acrylic inking plates and brayers with the scrub brush.
- Collect all sets of cutting tools and check that each one has all its pieces.
- Bring any unused linoleum blocks and test blocks back to the office.

Parent Instructions for Tidal Zone Prints

Thank you for volunteering today!

Please help by:

Setting up lesson supplies

- Clear desks/tables
- Pass out 1 sketch paper, 6 printing papers, paper towel, wooden spoon, 4" printing block, 1" x 2" printing block, and a set of carving tools
- Inking stations will be set up around the room or outside. Cover inking areas with paper. Place inking plates, brayers and inks, Q-tips, toothpicks and spray bottles on inking tables (inks to be squeezed out later when docent is ready)



During the lesson

- Listen carefully to the docent instructions to the students
- Assist individual students by walking around the room
- When students are sketching designs, encourage them to draw big, simple designs with texture, remind them that carving tools create thicker lines than pencils.
- Add ink to inking plates as needed by squeezing out no more than an inch of ink at a time
- Use up any open ink tubes first before opening new ones
- If inking plate is dry, spray plate with water to rehydrate
- Make sure students have their names on the back or under their print like a signature
- Remind students to bring their block back to their table to print, placing block down first and paper on top.
- Encourage students to place finished prints on drying rack (or where teacher requests)

Additional tips

- Please don't work on students' art — demonstrate on scrap materials
- Ask Art Docents if a student needs extra help
- Have fun!

Helping clean-up

- Wash inking plates and brayers (there is a wash tub and scrub brush in the unit)
- Wipe down tables

LOS ALTOS
art
DOCENTS



Background

KATSUSHIKA HOKUSAI 1760–1849 (Japanese)

Hokusai was born in Edo (now Tokyo) of unknown parentage, and was adopted into a mirror polisher's family; he began drawing at age 6. He could read and write—evidence of some education. Hokusai began training as a print designer at age 19 with a master of actor prints for a small Kabuki theater. Hokusai was a prolific artist creating over 30,000 designs. His most productive period began when the artist was in his seventies, when he produced his monumental series *Thirty-six Views of Mount Fuji* (actually 42) depicting people interacting with the natural environment. The *Great Wave* is the first in the series and is considered one of Hokusai's finest works. The three boats are *oshiokuri-bune* (fast boats that transported live fish up and down the coast). A giant wave is about to fall on helpless boatmen. While it seems perfectly balanced to Westerners—the wave would just roll on—to Japanese, accustomed to reading down and right to left, the wave was designed to tumble into the viewer's face. Hokusai used the balance of Japanese style and the influence of Western perspective, which makes the prints revolutionary to the Japanese public and appealing to Western eyes. The series was printed in shades of Prussian blue, an expensive and rare pigment recently introduced from the West.

TOM KILLION b. 1953 (American)

Tom Killion is an American artist, author, African historian and educator. He is internationally known for printmaking linocut, woodcut and letterpress techniques. The subject matter of his artwork is often the landscapes of Northern California.

KATSUSHIKA TAITO, FL. 1810–1853 (Japanese)

A prolific illustrator, Taito II was born in Edo to a samurai family from Kyushu. He began his artistic education under the great Hokusai, who bestowed his own name “Taito” onto his young student in 1820. Taito II worked closely with his teacher, collaborating with Hokusai on the second Manga volume in 1815. Working independently in Edo from 1830 to 1843, Taito II worked in Osaka between 1843 and 1853. Throughout his career, Taito II's oeuvre spanned *ehon* (illustrated books), single-sheet prints, and paintings. For a period of time, he began forging Hokusai's signature, but was found out. Following this incident, Taito II was sometimes called “*Inu* (Dog) Hokusai” or “Osaka Hokusai”

WAYNE THIEBAUD 1920-2021 (American)

An American painter best known for his still lifes of edible treats and everyday objects in his singular illustrative style. His most popular subject matter includes colorful cakes, slices of pie, candy pieces, and the winding streets of San Francisco. Born Morton Wayne Thiebaud on November 15, 1920 in Mesa, AZ, the painter moved to California soon after, where he became interested in stage design and lighting. He began his career as a commercial artist, but switched to fine art after during his education and earned an MFA from what is now the California State University at Sacramento. A trip to New York during the 1950s introduced him to the city's art scene at the time. Although he continued to show on the East Coast, Thiebaud spent most of his life in Sacramento.

Revision History

Revision Date	Docent Name	Comments
September 1999	Bonnie Sorenson	Lesson developed
February 2011	Kimberly Dickerson	Reformatted for continuity
February 2013	Curriculum Committee	Added suggestions for docents to give feedback to children
May 2015	Kimberly Dickerson	Added instructions for changing blades, changed transfer method, changed printing instructions to invert paper for rubbing, added spray bottle for re-hydrating ink, changed carving instructions to carve towards you, added scrub brush for clean-up
May 2019	Kimberly Dickerson	New VAPA Standards added
May 2022	Joan Takenaka	Clarify instructions for carving
February 2024	Kimberly Dickerson	Docent to bring in 6" x 9" paper for printing
June 2024	Curriculum Committee	Parent Volunteer Instructions added